2023-26 (Voc)

Full Marks : 70

Time : 3 hours

Answer from **both** the Sections as directed.

The figures in the right-hand margin indicate marks.

Candidates are required to give their answers in their own words as far as practicable.

SECTION–A

Answer any *four* questions: 10 × 4

1. Define group and subgroup. Show that a non-empty subset H of a group G is a subgroup if and only if $ab^{-1} \in H \ \forall \ a \in H, b \in H$.

2. Show that the set $C = \{a + bi : a, b \in \mathbb{R}\}$ of complex numbers is a field under usual addition and multiplication of complex numbers.

3. Define vector space and show that the set $M_2 = \left\{ \begin{bmatrix} a & b \\ c & d \end{bmatrix} : a,b,c,d \in \mathbb{R} \right\}$ is a vector space over $\mathbb{R}$ w.r.t. matrix addition and multiplication of a matrix by a scalar.

4. Show that $f : \mathbb{R}^4 \to \mathbb{R}^4$ given by $f(x,y,z,t) = (2x, 5y, 0, 0)$ is a linear transformation. Find its rank and nullity.

5. Solve the equation by matrix methods:
$3x + y + 2z = 3,\ 2x - 3y - z = -3,\ x + 2y + z = 4.$

6. Find the eigenvalues and eigenvectors of the matrix $\begin{bmatrix} 1 & 1 & 3 \\ 1 & 5 & 1 \\ 3 & 1 & 1 \end{bmatrix}$.

7. Find the magnitude and the equation of the shortest distance between the lines $\frac{x}{2} = \frac{y}{-3} = \frac{z}{1}$ and $\frac{x-2}{3} = \frac{y-1}{-5} = \frac{z+2}{2}$.

(*Continued*)

8. Find the equation of the sphere through $(0, 0, 0)$, $(0, 1, -1)$, $(-1, 2, 0)$ and $(1, 2, 3)$. Locate its centre and find the radius.

SECTION–B

3×10

9. Define Ring.

10. Define Boolean Algebra.

11. Show that $G = \left\{ \begin{bmatrix} a & 0 \\ 0 & 0 \end{bmatrix} : a \neq 0 \in \mathbb{R} \right\}$ is an abelian group under matrix multiplication.

12. Find the characteristic equation of the matrix $A = \begin{bmatrix} 1 & 1 & 3 \\ 1 & 3 & -3 \\ -2 & -4 & -4 \end{bmatrix}$ and hence its inverse.

13. Define quadratic forms with examples.

14. Show that the intersection of two subgroups of a group G is a subgroup of G.

15. Show that the set

$$W = \{(a_1, a_2, a_3) : a_1 + a_2 + a_3 = 0 : a_1, a_2, a_3 \in \mathbb{R}\}$$

is a subgroup of $\mathbb{R}^3$.

16. Define linear transformation and show that the mapping $T : \mathbb{R}^2 \to \mathbb{R}^2$ defined as $T(\alpha, \beta) = (\alpha, 0)$ is a linear transformations.

17. Define rectangular, spherical, polar & cylindrical co-ordinates.

18. Define direction cosines & direction ratios.

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Section—A

1. Answer any *four* questions: 10 × 4

   (*a*) Write and explain the structure and function of an ecosystem.

   (*b*) Write about nutrient flow biogeo-chemical cycle.

   (*c*) Write and explain the biological trans-formation of pollutants.

(*d*) Discuss:

(*i*) Human Rights

(*ii*) Woman and Child welfare

(*e*) Explain the classification of air pollutions.

(*f*) Write notes on :

(*i*) Toxic effects

(*ii*) Global Warming

(*g*) Write about Primary and Secondary waste water treatment.

(*h*) Write about characteristics of waste water.

Section—B

2. Answer *all* question: 3 × 10

(*a*) What is environmental science ?

(*b*) What is Ecosystem ?

(c) What is Ecological Pyramid?

(d) What is Ozone Depletion ?

(e) What is Food web ?

(f) What is Green House effect ?

(g) What is air pollution ?

(h) What is BOD ?

(i) What is Watershed management ?

(j) What is Water act ?

2023-26 (VOC)

Full Marks : 70

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Answer from **both** the Groups as directed.

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GROUP—A

1. Answer any *four* questions: 10 × 4

   (*a*) Explain the architecture of DBMS with all its components.

   (*b*) Explain the drawback of File based System.

   (*c*) What is a Key ? Write about different types of Keys.

(*d*) Write and explain BCNF with examples.

(*e*) Explain DDL and DML commands with examples.

(*f*) Explain Integrity Constraints and its types.

(*g*) What is a View ? What are the advantages of View over table ?

(*h*) Explain Group by, Having and order by command with examples.

GROUP—B

2. Answer *all* questions: 3 × 10

(*a*) Write about Constraints used in SQL.

(*b*) What is Nested Sub-queries ?

(*c*) What is a database management system ?

(*d*) What is the use of data dictionary ?

(*e*) What is the use of relational algebra ?

(*f*) Why is normalization needed in DBMS ?

(*g*) What is Generalization ? Explain with example.

(*h*) Whta is Relational model ?

(*i*) What are the different types of attributes ?

(*j*) What is the use of E-R Model ?

2023-26 (VOC)

Full Marks :70

Time : 3 hours

Answer from **both** the Sections as directed.

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SECTION–A

1. Answer any *four* questions: 10 × 4

(*a*) Explain the concept of object-oriented programming (OOP) and its fundamental principles.

(*b*) Describe the structure and syntax of a C++ program. Discuss the role of preprocessor directives, namespace, and main function with the help of an example.

(c) Explain the concept of classes and objects in C++. Discuss the syntax for defining classes, creating objects, and accessing class members with the help of an example.

(d) Describe the different types of constructors and destructors in C++. Discuss their purpose and usage in class initialization and cleanup. Write a program to show your explanation.

(e) Discuss the concept of inheritance in C++. Explain the different types of inheritance and their syntax. Describe the use of virtual functions and abstract classes.

(f) Explain the concept of operator overloading in C++. Discuss the syntax and usage of overloading unary and binary operators with the help of an example.

(g) Describe the concept of file handling in C++. Discuss the different modes of file operations and the syntax for opening, reading, writing and closing

files also write a program for it.

(*h*) Discuss the concept of exception handling in C++. Explain the try-catch block and the usage of different exception types.

SECTION–B

3 × 10

2. Answer *all* questions:

(*a*) What is the purpose of the using namespace std statement in C++ ?

(*b*) Explain the difference between call by value and call by reference in C++.

(*c*) Define the term "Inline Function" in C++.

(*d*) What is the purpose of the this pointer in C++ ?

(*e*) Explain the concept of function overloading in C++.

(*f*) What is the purpose of the friend function in C++ ?

(*g*) Define the term "Virtual Function" in C++.

(*h*) Explain the concept of pure virtual functions and abstract classes in C++.

(*i*) What is the purpose of the static keyword in C++ ?

(*j*) Explain the concept of templates in C++.

VUG(2)-BCA(C-2005)

2023-26 (VOC)

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Time : 3 hours

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GROUP—A

1. Answer any *four* questions: 10 × 4

(*a*) How does a J-K flip-flop differs from an S-R flip-flop in its basic operation. Explain.

(*b*) Differentiate between Synchronous Sequential Circuit and Asynchronous Sequential circuit. What do you mean by D- filp-flop ?

(*c*) What do you mean by ripple counters ? Explain 4 bit ripple counter and

also draw a timing diagram.

(d) Convert the following decimal numbers to the base indicated.

(*i*) 7562.45 to octal

(*ii*) 1938.257 to Hexadecimal

(*iii*) 175.175 to binary

(e) What do you mean by universal gate ? Show that both NAND gate and NOR gate are universal gates.

(f) Explain the k-map with three variables. Reduce the following function using K-map

$$F = B'D + A'BC' + AB'C + ABC'$$

(g) Design a full Subtractor circuit using Decoder and explain the working principle.

(h) How do you express the boolean function as a product of Maxterm ? Express the boolean function $F = XY + X'Z$ in a product of Maxterm form.

(Continued)

GROUP—B

2. Answer *all* question: (Compulsory) 3×10

(*a*) Draw the logic gates that implement the following:

$$F = (A+B)(B'+C)(C'+D+E)$$

(*b*) Define maxterm and minterm.

(*c*) Differentiate combinational logic and Sequential logic.

(*d*) What is flip-flop ? List the types of flip-flop.

(*e*) State and prove De-morgan's theorem.

(*f*) Explain what is an excitation table and what is a state table.

(*g*) What is the use of basic logic gates ?

(*h*) What is multiplexer ?

(*i*) Differentiate filp flop and latch.

(*j*) What is Shift register ?
